# Category Theory for Mathematical Physics — Exercise Sheet 0

*Diagnostic: what the course assumes*

**Variant: Questions only**

This optional sheet tests the prerequisites; each exercise names the sibling lectures that teach them and the lectures of this course that use them. Vector spaces are finite-dimensional over $\mathbb{C}$, and in $g \circ f$ the map $f$ acts first. For $V$ with basis $e_1, \dots, e_n$ ($n = \dim V$), the dual basis of $V^*$ is $e^1, \dots, e^n$, $e^i(e_j) = \delta_{ij}$; $m = \dim W$. $P$ is the flip $v \otimes w \mapsto w \otimes v$, and the partial trace $\mathrm{tr}_2 \colon \mathrm{End}(V \otimes W) \to \mathrm{End}(V)$ is $\mathrm{tr}_2(M) = \sum_k (\mathrm{id} \otimes e^k)\, M\, (\mathrm{id} \otimes e_k)$ for a basis $e_k$ of $W$. Groups are multiplicative, with identity 1. The singlet is written $|\psi^-\rangle$, and the antisymmetric tensor is $\Omega = e_1 \otimes e_2 - e_2 \otimes e_1$, as in Lecture 4.

**Exercise 1 (Duals and the trace).** (*Linear Algebra*, Lectures 11–12; used from Lecture 4.) Let $\mathrm{ev} \colon V^* \otimes V \to \mathbb{C}$, $\varphi \otimes v \mapsto \varphi(v)$, and $\mathrm{coev} \colon \mathbb{C} \to V \otimes V^*$, $1 \mapsto \sum_i e_i \otimes e^i$.

(i) Show that coev does not depend on the basis.

(ii) Show that $(\mathrm{id}_V \otimes \mathrm{ev}) \circ (\mathrm{coev} \otimes \mathrm{id}_V) = \mathrm{id}_V$ and $(\mathrm{ev} \otimes \mathrm{id}_{V^*}) \circ (\mathrm{id}_{V^*} \otimes \mathrm{coev}) = \mathrm{id}_{V^*}$.

(iii) Show that $\mathrm{ev} \circ P \circ (f \otimes \mathrm{id}_{V^*}) \circ \mathrm{coev} = \mathrm{tr}\, f$ for $f \colon V \to V$; in particular $\mathrm{ev} \circ P \circ \mathrm{coev} = \dim V$.

(iv) Show that $\varphi \otimes w \mapsto \big(v \mapsto \varphi(v)\, w\big)$ is an isomorphism $V^* \otimes W \to \mathrm{Hom}(V, W)$.

**Exercise 2 (Tensor products of maps).** (*Linear Algebra*, Lecture 12; *Quantum Mechanics I*, Lectures 10–11; used from Lecture 3.)

(i) Show that $(f \otimes g) \circ (f' \otimes g') = ff' \otimes gg'$ and $P \circ (f \otimes g) \circ P = g \otimes f$ for $f \colon V \to V$ and $g \colon W \to W$, where the right $P$ is the flip $W \otimes V \to V \otimes W$ and the left one $V \otimes W \to W \otimes V$.

(ii) On $\mathbb{C}^2 \otimes \mathbb{C}^2$, find the eigenvalues of $P$ with their eigenvectors.

(iii) Show that $P$ on $\mathbb{C}^2 \otimes \mathbb{C}^2$ is not of the form $f \otimes g$.

(iv) Show that $\mathrm{tr}_2(A \otimes B) = (\mathrm{tr}\, B)\, A$.

**Exercise 3 (Characters of $S_3$).** (*Symmetries in Physics*, Lectures 3–5 and 8; used from Lecture 2.) Let std be the permutation representation of $S_3$ on $\mathbb{C}^3$ minus its trivial summand.

(i) Write the characters of **1**, sgn and std on the three conjugacy classes, and check that they are orthonormal.

(ii) Decompose std $\otimes$ std.

(iii) Find $\dim \mathrm{Hom}_{S_3}(\mathrm{std} \otimes \mathrm{std}, \mathrm{std})$, and show that every $S_3$-map std $\to$ std is a scalar.

**Exercise 4 (Spin and Clebsch–Gordan).** (*Symmetries in Physics*, Lectures 12–14; *Quantum Mechanics I*, Lecture 13; used from Lecture 2, and in Lectures 4 and 9.) Let $V_{1/2} = \mathbb{C}^2$ be the defining representation of SU(2), with basis $e_1 = \uparrow$, $e_2 = \downarrow$, and $V_j$ the spin-$j$ representation.

(i) Show that $V_{1/2} \otimes V_{1/2} \cong V_0 \oplus V_1$, with $V_0$ spanned by the singlet $(e_1 \otimes e_2 - e_2 \otimes e_1)/\sqrt{2}$, and that the singlet is fixed by $U \otimes U$ for every $U \in \mathrm{SU}(2)$.

(ii) With the characters $\chi_j(t) = \sin\big((2j+1)t/2\big)/\sin(t/2)$ on the rotation by $t$ (extended continuously where the denominator vanishes), show that $\chi_{1/2}\chi_j = \chi_{j-1/2} + \chi_{j+1/2}$ for $j \geq \frac{1}{2}$, and deduce $V_{1/2} \otimes V_j \cong V_{j-1/2} \oplus V_{j+1/2}$.

(iii) Decompose $V_1 \otimes V_1$ and check the dimensions.

**Exercise 5 (Two qubits).** (*Quantum Mechanics I*, Lectures 10–11; used from Lecture 3.) Let $|\psi^-\rangle = (|01\rangle - |10\rangle)/\sqrt{2}$ in $\mathbb{C}^2 \otimes \mathbb{C}^2$, with $|0\rangle = e_1$ and $|1\rangle = e_2$.

(i) Show that $|\psi^-\rangle$ is not a product $|a\rangle \otimes |b\rangle$.

(ii) Compute the reduced state $\rho_1 = \mathrm{tr}_2 |\psi^-\rangle\langle\psi^-|$, and compare with $\mathrm{tr}_2$ of a product state.

(iii) How does $P$ act on $|\psi^-\rangle$, and on $(|01\rangle + |10\rangle)/\sqrt{2}$?

**Exercise 6 ⋆.** *An invariant cup with loop* $-2$ (a preview of Lecture 4). On $V = \mathbb{C}^2$ let $\Omega = e_1 \otimes e_2 - e_2 \otimes e_1 \in V \otimes V$, and let $\Omega^\dagger : V \otimes V \to \mathbb{C}$ be the linear map with $\Omega^\dagger(e_1 \otimes e_2) = 1$, $\Omega^\dagger(e_2 \otimes e_1) = -1$, $\Omega^\dagger(e_i \otimes e_i) = 0$.

(i) Show that $\Omega$ and $\Omega^\dagger$ are invariant under SU(2): $(U \otimes U)\Omega = \Omega$ and $\Omega^\dagger \circ (U \otimes U) = \Omega^\dagger$.

(ii) Compute $(\mathrm{id} \otimes \Omega^\dagger) \circ (\Omega \otimes \mathrm{id})$ and $(\Omega^\dagger \otimes \mathrm{id}) \circ (\mathrm{id} \otimes \Omega)$ as maps $V \to V$.

(iii) Show that if an invariant cup $\lambda\Omega$ and an invariant cap $\mu\Omega^\dagger$ satisfy both snake identities of Exercise 1, then the loop $(\mu\Omega^\dagger)(\lambda\Omega)$ is $-2$. Why can no invariant cup and cap that satisfy both snakes make the loop $+2 = \dim V$, as in Exercise 1?